

HUNTSVILLE, TX, USA

WMO: 722447

Lat: 30.744N

Lon: 95.586W

Elev: 112

StdP: 99.99

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53903

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-0.4	-9.7	1.7	3.9	-7.4	2.0	5.6	8.9	17.4	8.1	16.2	1.5	340	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.1	37.2	23.7	36.0	24.0	34.8	24.0	25.8	31.4	25.6	31.2	25.3	30.8	2.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	19.9	27.6	24.1	19.3	27.3	23.9	19.1	27.3	80.2	31.6	78.8	31.1	77.6	30.9	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
7.6	6.8	5.7	DB	-5.5	38.6	2.3	1.8	-7.1	39.9	-8.4	40.9	-9.7	41.9	-11.4	43.2	
			WB	-6.6	26.6	2.2	0.5	-8.1	27.0	-9.4	27.3	-10.6	27.5	-12.2	27.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.5	10.5	12.7	16.6	20.5	24.5	27.8	29.0	29.4	26.3	21.1	15.6	11.4
	DBStd	7.90	5.73	5.79	4.96	3.95	3.15	1.98	1.66	1.93	2.94	4.39	5.29	5.42
	HDD10.0	168	63	35	8	1	0	0	0	0	1	13	48	
	HDD18.3	899	249	170	92	24	2	0	0	0	1	24	113	224
	CDD10.0	4002	81	112	214	317	449	535	588	603	489	345	180	91
	CDD18.3	1690	7	13	39	90	193	285	329	344	239	110	31	9
	CDH23.3	17663	21	54	207	652	1789	3152	3928	4258	2444	962	165	31
Wind	WSAvg	2.3	2.7	2.9	2.8	2.7	2.5	2.2	2.1	1.9	1.5	1.9	2.2	2.5
	PrecAvg	1142	106	76	88	86	104	118	76	78	107	114	107	103
Precipitation	PrecMax	1550	292	182	179	237	325	385	179	190	241	459	360	214
	PrecMin	604	19	8	12	18	0	13	13	18	31	8	5	35
	PrecStd	240	64	47	43	54	76	82	47	55	49	103	78	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.8	27.4	28.9	31.8	34.5	37.6	37.7	39.1	37.6	33.0	28.7	26.2
		MCWB	18.4	18.0	19.6	22.2	23.2	23.3	23.5	23.9	22.5	22.6	21.6	20.4
	2%	DB	23.5	25.0	27.2	29.7	32.8	35.6	36.5	37.4	35.1	31.7	27.2	23.9
		MCWB	17.6	18.3	19.4	21.0	23.3	23.7	23.7	24.1	23.6	22.8	20.4	19.5
	5%	DB	22.0	23.0	25.9	28.3	31.9	34.0	35.2	36.2	33.6	30.1	25.3	22.4
		MCWB	17.7	17.9	18.9	20.5	23.1	24.1	24.1	24.3	23.1	21.6	19.2	18.8
	10%	DB	20.1	21.5	24.0	27.2	30.6	33.0	34.0	35.1	32.4	28.3	23.6	20.9
		MCWB	16.1	17.4	17.8	20.0	22.6	24.1	24.3	24.2	22.9	20.9	19.3	17.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.5	20.7	21.6	24.0	25.5	26.4	26.2	26.3	25.8	25.1	22.9	21.6
		MCDB	23.1	24.0	26.1	28.5	31.2	32.1	31.9	32.4	30.4	29.7	26.0	24.0
	2%	WB	19.4	20.1	20.8	22.9	24.7	25.7	25.7	25.8	25.4	24.4	21.9	20.6
		MCDB	22.3	23.1	24.6	27.2	30.0	31.1	31.4	31.8	30.0	28.3	25.0	23.0
	5%	WB	18.5	19.3	20.3	22.2	24.1	25.3	25.5	25.5	25.0	23.7	21.2	19.6
		MCDB	21.3	22.1	23.8	26.2	28.9	30.5	31.0	31.5	29.6	27.5	24.1	21.9
	10%	WB	16.9	18.3	19.6	21.6	23.6	24.9	25.2	25.2	24.5	22.9	20.2	18.2
		MCDB	19.4	20.8	22.8	25.4	28.3	30.1	30.6	31.1	29.1	26.6	23.1	20.2
Mean Daily Temperature Range	5% DB	MDBR	10.7	10.5	11.0	10.9	10.5	9.9	10.2	11.1	11.1	11.7	11.1	10.7
		MCDBR	11.3	10.9	10.9	11.1	10.9	11.1	11.7	12.3	12.3	11.7	11.2	11.3
		MCWBR	7.1	6.3	5.1	4.6	3.6	2.9	2.6	2.6	3.4	4.4	5.9	7.8
	5% WB	MCDBR	9.9	9.6	9.5	9.1	9.7	9.2	9.9	10.9	9.7	9.6	9.3	10.3
		MCWBR	7.6	7.0	6.0	4.9	4.0	3.0	2.8	2.7	3.2	4.5	6.2	8.7
Clear-Sky Solar Irradiance	taub		0.327	0.349	0.367	0.417	0.469	0.460	0.478	0.469	0.437	0.370	0.367	0.332
	taud		2.516	2.467	2.416	2.279	2.171	2.248	2.229	2.255	2.300	2.473	2.424	2.512
	Ebn at Noon		905	913	917	878	830	833	816	821	836	878	848	879
	Edh at Noon		88	101	113	135	151	139	141	136	126	99	96	84
All-Sky Solar Radiation	RadAvg		2.85	3.19	4.36	5.33	5.81	6.22	6.30	5.97	5.07	4.22	3.14	2.48
	RadStd		0.27	0.60	0.40	0.35	0.44	0.51	0.47	0.42	0.47	0.56	0.34	0.31

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	+93

Nomenclature: See separate page